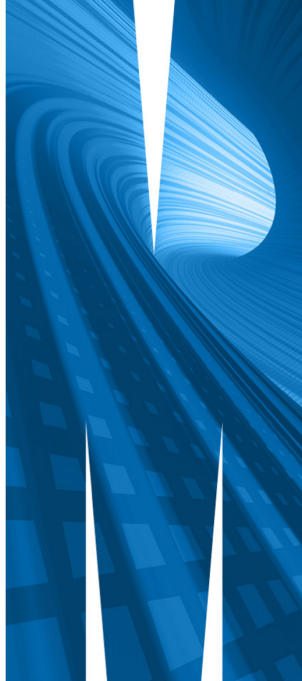


ETX1100/ETX5900

Business Statistics

Week 3: Sampling and estimation

8 December 2026



Outline

- 1 Samples and sampling
- 2 Sampling distributions
- 3 Confidence intervals for means
- 4 Confidence intervals for proportions
- 5 Reusable estimation workflow

Today's journey

- 1 Connect population questions to random samples.
- 2 Recognise bias and sampling variation.
- 3 Use the law of large numbers and central limit theorem.
- 4 Build confidence intervals for means and proportions.
- 5 Explain precision, confidence, and limitations in business language.

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Retrieval check

- What is the difference between a parameter and a statistic?
- What does $P(A \mid B)$ do to the denominator?
- When is a t distribution useful?
- Why does a large dataset not automatically represent its target population?

Population to sample and back

- 1 Define the **target population**.
- 2 Select a sample using a defensible process.
- 3 Calculate a sample statistic.
- 4 Use the statistic to estimate a population parameter.
- 5 Quantify uncertainty and communicate limitations.

Inference is only as credible as both the sampling process and the analysis.

Notation for means

Population parameter	Sample statistic
mean μ	mean $\bar{x}$
standard deviation σ	standard deviation s

Greek letters describe the population; Roman letters describe the observed sample.

Notation for proportions

- Population proportion: π .
- Sample count with the characteristic: X .
- Sample size: n .
- Sample proportion:

$$\hat{p} = \frac{X}{n}.$$

A proportion must lie between 0 and 1.

Sampling bias

Bias is a systematic tendency to over- or under-represent the target quantity.

Common sources include:

- undercoverage;
- voluntary response;
- non-response;
- convenience sampling; and
- leading or ambiguous measurement.

Increasing n reduces random variation, not systematic bias.

Watch me: diagnose a survey

An app asks active users, “How satisfied are you with our excellent new design?”

Identify:

- target population;
- sampling frame;
- likely missing users;
- wording bias; and
- a more credible design.

Now you try: redesign the evidence

Time allowed: 6 minutes

A smartwatch company emails only customers who completed a workout yesterday and asks whether its product increases exercise.

- 1 Identify two sources of bias.
- 2 Explain why a very large response does not fix them.
- 3 Propose a sampling and measurement plan.
- 4 State the population to which your conclusion would apply.

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Statistics vary from sample to sample

If we repeatedly take random samples of the same size, we get different values of $\bar{x}$ and $\hat{p}$.

A **sampling distribution** describes those possible statistic values—not the distribution of individual observations.

This sampling variation is the uncertainty quantified by standard errors and confidence intervals.

Law of large numbers

As a random sample grows, the sample mean tends to settle closer to the population mean.

- It does not say every larger sample is closer.
- It does not repair biased sampling.
- It describes long-run behaviour under repeated random sampling.

Central limit theorem

For sufficiently large random samples, the sampling distribution of the mean is approximately Normal:

$$\bar{X} \approx N\left(\mu, \frac{\sigma}{\sqrt{n}}\right).$$

The approximation concerns **sample means**, even when individual observations are not Normal.

Standard error of a mean

The standard deviation of the sampling distribution is the standard error:

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} \quad \text{or estimated by} \quad \frac{s}{\sqrt{n}}.$$

As n quadruples, the standard error halves. Precision improves with the square root of sample size.

Watch me: compare sample sizes

Suppose $\sigma = 20$.

- At $n = 25$, $SE = 20/\sqrt{25} = 4$.
- At $n = 100$, $SE = 20/\sqrt{100} = 2$.

The population variability did not change; uncertainty in the sample mean did.

Now you try: sampling variation

Time allowed: 5 minutes

Assume transaction values have $\sigma = 48$.

- 1 Calculate $SE(\bar{X})$ for $n = 36$ and $n = 144$.
- 2 Which sample mean is more precise?
- 3 How much must n change to halve the standard error?
- 4 Would either calculation remove non-response bias?

Sampling distribution of a proportion

For a sufficiently large random sample:

$$\hat{p} \approx N \left(\pi, \sqrt{\frac{\pi(1 - \pi)}{n}} \right).$$

For estimation, check using the observed proportion:

$$n\hat{p} \geq 5, \quad n(1 - \hat{p}) \geq 5.$$

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Point estimate plus margin of error

Every confidence interval has the same architecture:

estimate $\pm$ critical value $\times$ standard error.

- The point estimate identifies the centre.
- The margin of error quantifies sampling uncertainty.
- The confidence level controls the long-run coverage rate.

Known versus unknown σ

For a population mean:

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}} \quad (\sigma \text{ known}),$$

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}} \quad (\sigma \text{ unknown, } df = n - 1).$$

In business applications, σ is usually unknown, so the t interval is common.

Critical values in Excel

For confidence level $1 - \alpha$:

- $z^* = \text{NORM.S.INV}(1 - \alpha/2)$
- $t^* = \text{T.INV}(1 - \alpha/2, n - 1)$

For 95% confidence, $z^* \approx 1.96$. A t^* value is larger for small samples and approaches 1.96 as n grows.

Watch me: audit balances, known σ

An audit sample has $n = 80$, $\bar{x} = 86.05$, and known $\sigma = 22.38$.

In Excel:

- 1 compute =NORM.S.INV(0.975);
- 2 compute =22.38/SQRT(80);
- 3 multiply for the margin of error; and
- 4 calculate lower and upper bounds.

Interpret the interval for the **population mean balance**.

Now you try: audit balances, unknown σ

Time allowed: 8 minutes

Use $n = 80$, $\bar{x} = 86.05$, and $s = 24.02$.

- 1 Construct a 95% t interval.
- 2 Identify df and the Excel critical-value formula.
- 3 State the margin of error.
- 4 Interpret the interval in dollars and in context.

Interpreting confidence correctly

A 95% confidence method captures the fixed population parameter in about 95% of repeated random samples.

For the interval already calculated, say:

We are 95% confident that the population mean lies between the calculated bounds.

Do not say 95% of individual observations lie inside the interval.

Now you try: confidence versus precision

Time allowed: 6 minutes

Repeat the audit interval at 90% and 99% confidence.

- 1 Rank the three intervals by width.
- 2 Explain the confidence–precision trade-off.
- 3 Predict what happens if n decreases.
- 4 Identify one aspect of data quality that interval width does not capture.

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Interval for a population proportion

When the large-count condition is satisfied:

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}.$$

The standard error uses $\hat{p}$ because the unknown π is the quantity being estimated.

Watch me: recruitment-service use

Of 600 surveyed employers, 126 used a recruitment service recently.

- 1 $\hat{p} = 126/600 = 0.21$.
- 2 Check $600(0.21)$ and $600(0.79)$.
- 3 Use $z^* = 1.96$ for 95% confidence.
- 4 Calculate the standard error and bounds.
- 5 Format the result as percentages.

Now you try: product trial proportion

Time allowed: 8 minutes

Open `week3-proportions-data.xlsx`, sheet **Marketing**.

- 1 Find n , the count who have tried the product, and $\hat{p}$.
- 2 Check the large-count condition.
- 3 Construct a 95% confidence interval for π .
- 4 Does the interval suggest that more than half the market has tried it?
- 5 State a conclusion and one sampling limitation.

Compare groups carefully

Separate confidence intervals can describe subgroup proportions, but visual overlap is not a formal test of a difference.

Before comparing groups, ask:

- Were both groups sampled credibly?
- Are sample sizes adequate?
- Are categories defined consistently?
- Is the practical difference meaningful?

Now you try: subgroup intervals

Time allowed: 9 minutes

For female and male respondents separately:

- 1 calculate n , X , and $\hat{p}$ for product trial;
- 2 construct a 95% interval for each population proportion;
- 3 compare centres and widths; and
- 4 write a cautious conclusion without claiming causation.

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Watch me: build an Excel template

Create labelled input cells for:

- estimate, standard deviation or count, n , and confidence level;
- critical value and standard error;
- margin of error; and
- lower and upper bounds.

Keep inputs visually separate from formulas. Add an assumption check and a plain-language conclusion beneath the calculations.

Now you try: scenario analysis

Time allowed: 8 minutes

Using your template, hold the estimate fixed and compare:

- 1 90%, 95%, and 99% confidence;
- 2 sample sizes $n = 50, 200$, and 800 ; and
- 3 mean versus proportion standard errors.

Write two general rules a manager could use when planning a study.

Common traps

- Confusing the distribution of observations with the sampling distribution.
- Assuming large n removes selection or measurement bias.
- Using z when σ is unknown without justification.
- Forgetting the square root of n .
- Interpreting a confidence interval as containing most individual values.
- Reporting bounds without units, population, or business meaning.

Closing check

Can you:

- distinguish bias from random sampling variation?
- explain the law of large numbers and central limit theorem?
- calculate and interpret a standard error?
- choose a mean or proportion interval and its critical value?
- explain how confidence level and sample size affect width?
- communicate an interval as evidence, not certainty?